BIOMEDICAL LIMITED, ILORIN, No 1 Ohimege Road, Industrial Estate, Ilorin Kwara State, PMB 1499.	RAW MATERIAL ANALYTICAL SPECIFICATIONS & METHODS	STP №	: BML/QA/STP/RMA/009/F02
		Material Code №	: RMA009
		Effective Date	: 4th Jan., 2024
		Supersedes	:
		Review Date	: 4th Dec., 2026
	MANNITOL	Revision No	: 01
		Ref. Book	BP 2023/IN-HOUSE
		Page	: 1 Of 11
Storage Conditions:	**Store in a well-closed container, protected from light**	**Department**	**: Quality**
Retest Frequency:	**12 months from date of release**	**Section**	**Quality Control**

ANALYTICAL METHODS

1 DESCRIPTION

1.1 Physically examine a representative sample of the material for Appearance and other characteristics: it A white, or almost white crystalline powder, freely soluble in water, very slightly soluble in ethanol, sample shows polymorphism.

2 IDENTIFICATION

2.1 **Melting Range**

2.1.1 Determine the melting point using the capillary method: **It melts at between 165 - 170°C**

2.2 Catechol-Sulphuric Acid Test

2.2.1 10% w/v solution of catechol

2.2.1.1 Accurately weigh and transfer about 5g of sample into a 50ml volumetric flask

2.2.1.2 Add 30ml of distilled water and mix.

2.2.1.3 Mechanically shake till dissolution is complete.

2.2.1.4 Dilute to the mark with distilled water and mix.

2.2.2 Sample preparation

2.2.2.1 Accurately weigh and transfer about 25mg of sample into a 50ml volumetric flask.

2.2.2.2 Add 30ml of distilled water and mix.

2.2.2.3 Mechanically shake till dissolution is complete.

2.2.2.4 Dilute to the mark with distilled water and mix.

2.2.3 procedure

2.3.3.1 Pipette 5ml of sample in a test tube.

2.3.3.2 Add to the sample, a mixture of 6ml of sulphuric acid and 3ml of freshly prepared 10% w/v solution of catechol. cool in ice

2.3.3.3 Heat gently over a naked flame for about 30 seconds.

2.3.3.4 A pink colour develops

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

3 SULPHATED ASH

3.1 Weigh a crucible (W1) that has previously been ignited and cooled.

3.2 Accurately weighs about 1g onto each crucible, ensuring they are evenly spread, then determine their weight together (W2).

3.3 Heat, gently at first on a Bunsen flame, until the sample is thoroughly charred.

3.4 Cool, then moisten the residue with 1mL of sulphuric acid, heat gently until white fumes are no longer evolved, and ignite at 800 + 25oC until the carbon is consumed (about 2 hours).

3.5 Cool in a desiccator, weigh, and calculate the % ROI, if the %ROI exceed 0.1%, again moisten the residue with 1mL of sulphuric acid, heat and ignite as before.

3.6 Continue until constant weight is attained (until two successive weighing does not differ by more than 0.5mg).

3.7 Weigh the crucibles with its content (W3).
Calculate the Sulphated Ash content as follows:

$$\% \text{ Sulphated Ash} = \frac{(W3 - W1) \times 100}{W2 - W1}$$

Where,
W1 = weight of empty crucible, g
W2 = weight of sample and crucible, g.
W3 = weight of crucible and the dried sample, g

4 REDUCING SUGAR TEST

4.1 To 7.0 g add 13 mL of water R. Boil gently with 40 mL of cupri- tartaric solution R for 3 min, and allow to stand for 2 min. A precipitate is formed

4.2 Filter through a sintered-glass filter (16) (2.1.2) coated with diatomaceous earth R or a sintered-glass filter (10) (2.1.2).

4.3 Wash the precipitate with hot water(about 50-60°C

4.4 until the washing is no longer alkaline

4.5 and filter the washings through the same sintered glass filter

4.6 Discard the filtrate.

4.7 Immediately dissolve the precipitate in 20 mL of ferric sulfate solution R filter through

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

the same sintered-glass filter

4.8 and wash the filter with15-20 mL of water R. Combine the washings and the filtrate, heat to 80 "C,

4.9 titrate with 0.02 M potassium permanganate .Not more than3.2 mL is required [0 change the colour of the solution from green to pink so that the colour persists for at least 10 s.

5 SPECIFIC OPTICAL ROTATION

5.1 Accurately weigh and transfer 2g of sample into a 25ml volumetric flask.

5.2 Add 2.6g of sodium tetraborate, mechanically shake/sonicate till dissolution is complete.
Dilute to the mark with distilled water and mix.

5.3 allow standing for 30 minutes.

5.4 Using a suitable polarimeter and polarimeter tube, measure the optical rotation of the blank solution prepared by weighing 2.6g of sodium tetraborate to 100.0ml with distilled water

5.5 Inject the blank at 20°C and zero the instrument.

5.6 Inject the prepared sample solution and observe the optical rotation at 20°C, making at least 5 readings and determine the average.

5.7 **Calculation:**

$$[\alpha]^{25} = \frac{100\ (\alpha)}{\text{Length Of Polarimeter tube (dm) X Sample Weight (g)}}$$

Where: $[\alpha]^{25}$ = specific rotation

$\alpha 1$ = observed rotation of sample preparation at temperature 20°C

.

6 CONDUCTIVITY

6.1 Dissolved 20g of sample in carbon dioxide – free water R

6.2 Prepared from distilled water R, by heating at 40-50°C and dilute to 100ml with the carbon dioxide free water.

6.3 Allow the solution to cool

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

6.4 Measure the conductivity of the sample,

6.5 The conductivity should Not More Than 20 $\mu S \cdot cm^{-1}$.

7 LOSS ON DRYING

7.1 Weigh one labelled petri dish (W1) that has previously been dried at 105°C and cooled.

7.2 Accurately weigh about 1g onto a petri-dish, ensuring they are evenly spread, then determine their weight together (W2).

7.3 Transfer into an Oven operating at 105°C.

7.4 Dry for 4 hours

7.5 Transfer into a desiccators to cool to room temperature.

7.6 Weigh the petri-dish with its content (W3

7.7 Calculate the moisture content as follows:

$$\% \text{ Moisture} = \frac{(W2 - W3) \times 100}{(W2 - W1)}$$

Where,

W1 = weight of empty petri-dish, g.

W2 = weight of sample and petri-dish, g.

W3 = weight of petri-dish and the dried sample, g.

8 Heavy metals

8.1 Lead nitrate stock solution(100ppm)

8.1.1 Dissolve 160mg of lead nitrate, Analar, in 100ml distilled water to which has added 1ml nitric acid,

8.1.2 Dilute with distilled water to 1000ml and mix. Prepare and store this solution in glass containers free from soluble lead salts.

8.2 Lead Nitrate Stock Solution (1ppm)

8.2.1 Pipette1.0ml of the 100ppm lead standard solution into a 100ml volumetric, dilute to the mark with distilled water and mix.

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

8.3 buffer solution ph3.5

8.3.1 In a 250ml beaker, dissolve 25.0g of ammonium acetate in 25ml of water and add 38.0ml of 7m hydrochloric acid.

8.3.2 Adjust the ph if necessary with 2m hydrochloric acid or 6m ammonium hydroxide and dilute to 100.0ml with water.

8.4 Thioacetamide solution (4%)

8.4.1 Dissolve 4g of thioacetamide in a 100ml volumetric flask, make up with distilled water

8.4.2 Thioacetamide Reagent

8.4.3 To 2ml of thioacetamide solution in a 100ml volumetric flask, add 7.5ml of water, 37.5ml of M sodium hydroxide and 50ml of glycerol (85%). Heat in a water-bath for 20 seconds. Prepare immediately before use.

8.5 Sample Preparation

8.5.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

8.5.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

8.5.3 Dilute to the mark with same solvent and mix.

8.5.4 To 12ml of the solution thus obtained (in a test tube), add 2ml of buffer solution ph3.5, 1.2ml of thioacetamide reagent and immediately mix.

8.6 Standard Preparation

8.6.1 Transfer 10 ml of 1ppm lead standard solution and 2 ml of the sample solution into a suitable test tube.

8.6.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

8.7 Blank Preparation

8.7.1 Prepare a blank using a mixture of 10ml of water and 2ml of sample solution.

8.7.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

8.8 OBSERVATION

8.8.1 Compared to the blank, after 2 minutes any brown colour in the solution obtained with the sample is not more intense than in the standard solution

9 **ASSAY**

9.1 **Method A**

9.1.1 **Reagents**

9.1.1.1 **5.3% sodium carbonate**

Weight 5.3g of sodium carbonate in 100ml volumetric flask mark up to mark with distilled water

9.1.2 **SODIUM THIOSULFATE, (0.1 N)**

9.1.2.1 Dissolve about 26g of sodium thiosulfate and 200mg of sodium carbonate in 1000ml of recently boiled and cooled water. Standardize the solution as follows

9.1.2.2 Weigh accurately about 210mg of primary standard potassium dichromate, previously pulverized and dried at 120°C for 4 hours, and dissolve in 100ml of water in a glass-stoppered, 500-ml flask

9.1.2.3 Swirl to dissolve the solid, remove the stopper, and quickly add 3g of potassium iodide, 2g of sodium bicarbonate, and 5ml of hydrochloric acid. Insert the stopper gently in the flask, swirl to mix, and allow standing in the dark for 10 minutes. Rinse the stopper and the inner walls of the flask with water

9.1.2.4 titrate the liberated iodine with the sodium thiosulfate solution until the solution is yellowish green in color. Add 3ml of starch TS, and continue the titration to the discharge of the blue color. Re-standardized the solution frequently

$$\text{(Molarity) of } Na_2S_2O_3 = \frac{\text{Weight of Potassium dichromate (mg)} \times 6}{294.2 \times \text{Titre}}$$

9.1.3 **0.05M (0.1N) Iodine**

Dissolve about 14g of iodine in a solution of 36g of potassium iodide in 100ml of water, add 3 drops of hydrochloric acid, dilute with water to 1000ml

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

9.1.4 **15%v/v sulphuric acid**
Measure 75mL of concentrated sulphuric acid into 500mL volumetric flask and make to the mark with distilled water

9.1.5 2%w/v starch mucilage
To 2g of soluble starch add 100ml of boiling water and boil for 1 minute with continuous stirring and Cool.

9.1.6 Sample Preparation
9.1.6.1 Accurately weigh about 2g of sample into a 500ml volumetric flask.
9.1.6.2 Add 200ml of distilled water, mechanically shake/sonicate till dissolution is complete
9.1.6.3 Dilute to the mark with distilled water and mix

9.1.7 Procedure
9.1.7.1 Pipette 5.0ml each of the sample solution and distilled water (blank) into separate 250ml Iodine flasks, add 10.0ml of 5.3%w/v sodium carbonate solution, 25.0ml of 0.05M (0.1N) Iodine solution and mix.
9.1.7.2 Allow standing in the dark for 15 minutes, adding 100ml of 15%v/v sulphuric acid and mix
9.1.7.3 Titrate the excess iodine with 0.1M (0.1N) sodium thiosulphate to a faint yellow colour.
9.1.7.4 Add 2ml of 2%w/v starch mucilage and continue titration till the blue-black colour of the starch-Iodine complex is discharged.

$$\%Assay = \frac{(\text{Blank titre- Sample titre}) \times \text{Actual Molarity of } Na_2S_2O_3 \times 90080}{\text{Sample weight (g)} \times (100\text{-\%Loss On Drying})}$$

10 **Microbiological Tests**

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

10.1 **Limulus Amebocyte Lysate LAL Test**

10.1.1 Open the seals of the test tubes of lyophilized LAL Reagents

10.1.2 Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes.

10.1.3 Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam.

NOTE: cover the reaction tubes with pyrogen free aluminium foil

10.1.4 Incubate the reaction mixture immediately at 37Oc for 60minutes. Avoiding vibration during incubation.

10.1.5 If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit.

10.1.6 Sample to be tested should be stored under conditions that bacteriological activities are controlled

10.1.7 Read reaction tubes one by one.

10.1.8 Invert the tube in one smooth motion.

10.1.9 A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. A negative result is characterized by the absence of solid clot after inversion.

Testing Methodology of Bacterial Endotoxin (LAL)

10.2.1 Each specimen shall be tested in duplicate

10.2.2 NMC: Take 100µl(0.1ml) LRW

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

10.2.3 Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution.

10.2.4 Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

10.2.5 Add the LAL reagents as mentioned below in the table 1

10.2.6 Mix gently and incubate the tubes in heating block at 370C + 1 for 60min.

10.2.7 After incubation, remove the tubes gently from the heating block and slowly invert through 1800C and note the observations.

10.2.8 If a firm gel has formed, record the results as positive.

10.2.9 A result is negative, if an intact gel is not formed.

10.2.10 Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material.

10.3 **Pyrogen**

10.3.1 Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed.

10.3.2 Withhold all food from the rabbits used during the period of the test

10.3.3 Draw out the required number of rabbits from their living room according to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring.

10.3.4 All the rabbits which are to be used for the test should be more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit.

10.3.5 Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm.

10.3.6 If the initial temperature is less than 38.80c or more than 39.8ocexclude the

animal from the test.

10.3.7 Ensure all the needles and syringe to be used are new

10.3.8 Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 10c.

10.3.9 Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection.

10.3.10 Any rabbits showing more than 0.20c temperature difference between two successive readings in the last three readings can be excluded from the test.

10.3.11 Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes.

10.3.12 Inject the solution intended to be tested to the marginal veins of the rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution.

10.3.13 If the substance to be tested is not in aqueous sterilized solution form, consult the official monograph regarding the preparation of simulating solution and the volume of the preparation to be injected.

10.3.14 After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the injection.

10.3.15 The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution

10.4 **MLT**

10.4.1 Weigh 1g or 1ml of the raw material sample and dissolve aseptically in 9ml sterile soybean casein digest medium prepared as per SOP for media preparation (BML-QDG-MC-011).

10.4.2 Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method.

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

10.4.3 Testing to be performed under LAF of specified area

10.4.4 Arrange all the required material on the LAF before starting the test.

10.4.5 Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes

10.4.6 Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP(BML-QDG-MC-011), (cooled approximately up to 45oC) in two petri plates for TAMC

10.4.7 Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45oC) in two petri- plates for TYMC

10.4.8 Gently swirl the contents inside the petri-dishes for proper mixing of the sample, allow medium to solidify, taking the precaution of not spilling the media from the plate

10.4.9 Simultaneously perform the negative control by plating 1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media previously maintained not more than 45Oc and allow for solidification.

10.4.10 Incubate the plates at 30oC to 35OC for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 200C to 25oC for 5 days in an inverted position.

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		